

LAB 4 - VIBECODING

Dino Run - Chat / LLM History

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SRN	PES1UG24AM219
LLM used	ChatGPT
Dino Run repository	Ramalakshmi-1205/36_dinoRun
Lab submission repository	Ramalakshmi-1205/SE_LAB1_PES1UG24AM219

About this document

This is my ChatGPT working history for Lab 4. I used ChatGPT to understand the Dino Run README, inspect the existing game, fix the broken high-score logic, add the three required features one by one, test the changes, and complete the GitHub work. I have kept the explanations short and in the order in which I completed the work.

1. Starting the project

I forked the assigned Dino Run repository, cloned my fork, inspected the README, and ran the original Pygame game before making changes. I recorded the required BEFORE video before starting the feature changes. After completing the code, I copied the updated game.py into my personal lab repository under the Lab-4 folder.

My prompt:

I have this VibeCoding Dino Run lab. Tell me what I need to do and guide me step by step.

ChatGPT guidance:

The main steps were to read the README, fork or clone the repository, run the original game, record the BEFORE video, fix the broken code, add the listed features one by one, test each change, record the AFTER video, make separate commits, and push the work to my personal repository.

What I did: The original game ran successfully and I recorded my BEFORE video before making the changes.

2. Task 1 - Fix High-Score Bug

My prompt:

Fix the broken high-score logic in the Dino Run game. Make sure the current score replaces the saved high score only when the current score is higher.

ChatGPT guidance:

The comparison was reversed. The current run score should be compared with the saved high score. If the current score is greater, the high score should be updated and saved.

Code used:

```
if current > self.high_score:
    self.high_score = current
    save_high_score(self.high_score)
```

Testing: I ran the game and checked that the high-score behaviour worked correctly. I then committed and pushed the change.

Commit message: Fix high score update bug

3. Task 2 - Dino Jump Tint

My prompt:

Implement the dino_tint feature. Make the Dino look different while it is jumping, but keep the normal colour when it is on the ground.

ChatGPT guidance:

The function receives on_ground. It can return None when the Dino is on the ground and an RGB colour when the Dino is airborne. The existing Dino.draw method already uses this return value.

Code used:

```
def dino_tint(on_ground):  
    if on_ground:  
        return None  
    return (255, 165, 0)
```

Testing: I ran the game and confirmed that the Dino stayed green on the ground and became orange while jumping.

Commit message: Add dino jump tint

4. Task 3 - Obstacle-Pass Popup

My prompt:

Implement on_obstacle_passed. I want a simple visible indication when the Dino successfully passes an obstacle.

ChatGPT guidance:

A simple +1 popup can be used. The callback returns "+1" when an obstacle is passed. The game stores the popup text for a short time and draws it above the Dino.

Code used:

```
def on_obstacle_passed(obstacle, score):  
    return "+1"
```

Additional change: Popup text and a short timer were added to the Game state so that the message appears briefly after an obstacle is passed.

Testing: I ran the game and confirmed that the +1 popup appeared after successfully passing an obstacle.

Commit message: Add obstacle pass popup

5. Task 4 - Double Jump

My prompt:

Implement max_jumps so that the Dino can perform a double jump without changing the rest of the jump behaviour.

ChatGPT guidance:

The existing jump logic checks max_jumps(). Returning 2 allows the Dino to jump twice before it must land again.

Code used:

```
def max_jumps():  
    return 2
```

Testing: I pressed Space once and then again while the Dino was still in the air. The Dino successfully performed two jumps.

Commit message: Add double jump

6. GitHub / Version Control Work

For the lab, I worked on my own fork of the assigned repository and pushed the changes there. I kept separate commits for the tasks and did not raise a pull request to the original SETAPESU26 repository.

Repository workflow:

SETAPESU26/36_dinoRun → Fork → Ramalakshmi-1205/36_dinoRun → Task changes → Separate commits → Push

Relevant commits: Fix high score update bug; Add dino jump tint; Add obstacle pass popup; Add double jump.

For the final Lab-4 submission, I uploaded the updated game.py, BEFORE video, AFTER video, and this ChatGPT working-history PDF to the Lab-4 folder of my personal lab submission repository.

7. Final Testing

Feature	What I checked
High score	Current run score updates and saves when it beats the previous best.
Dino tint	Dino is green on the ground and orange while jumping.
Obstacle popup	A +1 popup appears after an obstacle is successfully passed.
Double jump	Dino can perform two jumps before landing.
Gameplay	The original start, movement, obstacle collision, score, and restart behaviour still work.

8. Final Video

BEFORE video: PES1UG24AM219_DinoRun_Before.mp4

AFTER video: PES1UG24AM219_DinoRun_After.mp4

I recorded the BEFORE video before making the feature changes and the AFTER video after completing and testing the required tasks. The AFTER video shows the updated Dino Run game with the implemented features working.

9. Submission

The final Lab-4 folder contains the updated game code, the BEFORE and AFTER videos, and this ChatGPT working-history PDF. The updated code is also available in my personal fork of the assigned Dino Run repository.